

TALHA GHAFOR

AI/ML Engineer

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PROFESSIONAL SUMMARY

Machine Learning Engineer / AI Engineer — Software Engineering graduate (HITEC University, 2026) with hands-on experience in machine learning, deep learning, and NLP, including transformer-based models, computer vision, and model evaluation/optimization (precision, recall, F1-score, AUC-ROC, SMOTE, hyperparameter tuning). Designed and shipped the model-serving and inference architecture for a full-stack, multi-modal generative AI platform (FYP), including quantized model deployment on AWS. Core stack: Python, PyTorch, scikit-learn, HuggingFace Transformers, FastAPI, AWS, Docker, Git.

PROFESSIONAL EXPERIENCE

AI and Machine Learning Engineering Intern Jul 2025 – Sep 2025

Digital Empowerment Network

- Built and tuned an NLP mental health detection model (TF-IDF + Logistic Regression), improving performance to ~84% F1-score through iterative threshold tuning and feature refinement.
- Diagnosed a class-imbalance issue in a heart disease detection pipeline and applied SMOTE resampling, raising minority-class recall from ~71% to 83%.
- Authored model evaluation reports (accuracy, precision, recall, AUC-ROC) and presented findings to the engineering team at project close-out.

AI and Machine Learning Engineering Intern Jun 2025 – Jul 2025

DevelopersHub Corporation

- Trained and validated a CNN-based image classifier across multiple iterations, reaching ~89% validation accuracy against target benchmarks.
- Cleaned, labeled, and preprocessed a ~6,000-image dataset to reduce overfitting and improve data quality ahead of training.
- Documented training results and error patterns across runs in weekly reports, coordinating with the ML team to prioritize fixes.

KEY PROJECTS

GenMark — Multi-Modal AI Content Generation Platform (Final Year Project) ([Live Demo](#)) 2025 – 2026

HITEC University | Python, PyTorch, FastAPI, React, AWS, Docker

- Fine-tuned and optimized Gemma-2-9B, FLUX.1-schnell, and Qwen2-VL-7B models for domain-specific content generation, then deployed them with 4-bit quantization across dual T4 GPUs.
- Deployed backend infrastructure via AWS ECS Fargate, Lambda, and API Gateway, with a Flask, MongoDB, Celery, and Redis service layer.
- Built the React/TypeScript frontend with a Three.js interface, integrating real-time content generation and preview.

Hybrid Transformer-Affective Mental Health Detection ([GitHub](#)) 2025

Personal Project | Python, PyTorch, HuggingFace Transformers, scikit-learn, SHAP

- Designed a hybrid deep learning architecture fusing RoBERTa semantic embeddings with sentiment and emotion feature vectors (778-dim) to classify ~10,000 labeled Reddit posts into 4 mental health categories, extending a published binary-classification baseline (Kerasiotis et al., 2024) to a harder multiclass task; used class-weighted loss to handle class imbalance.
- Achieved 83.83% accuracy and 83.73% macro F1-score (94% F1 on the Anxious class), matching the baseline's accuracy on a significantly harder 4-class task.
- Implemented SHAP-based explainability to surface which words drove each prediction, supporting model transparency for a sensitive-domain use case.

TECHNICAL SKILLS

AI & ML: Machine Learning, Deep Learning, PyTorch, scikit-learn, HuggingFace Transformers, NLP, Computer Vision, CNN, RoBERTa/BERT, Stable Diffusion, Qwen, Model Deployment, Data Preprocessing, Feature Engineering, Model Evaluation & Optimization (Precision/Recall/F1/AUC-ROC, Hyperparameter Tuning), SMOTE, Explainable AI (SHAP)

Core Languages: Python, JavaScript

Web & Backend: FastAPI, Flask, React, REST APIs

Databases & Cloud: PostgreSQL, MongoDB, MySQL, AWS (ECS Fargate, Lambda, API Gateway, S3), Docker

Tools: Git, GitHub, Jupyter Notebook, VS Code

EDUCATION

BS Software Engineering — Graduated 2026

HITEC University, Taxila — Dept. of Computer Science

Intermediate – ICS, KRL College, Kahuta (2020–2022) | Matriculation – CS, Mumtaz Grammar School, Wazirabad (2018–2020)

CERTIFICATIONS & LEADERSHIP

OCI 2025 Certified AI Foundations Associate, Oracle University (Aug 2025 – Aug 2027) | Certified Python & Machine Learning, Mars Computer Academy (2025)

Security Team Head, HITEC Olympiad (2024) | Assistant Head, CodeWar 2026